10th April 2026

To whom it may concern

Subject: Recommendation for Ryan Bantu

I am writing this letter as a recommendation for Ryan Bantu, who was a member of Team Kalpana from the Manipal Institute of Technology. The team was awarded first prize in the Residential Cooling Retrofit (RCR) division of the Solar Decathlon India (SDI) 2024–25 competition.

During the competition, Ryan played a key role in the development of the team's innovative solution, VAYU, a retrofit cooling device designed for hot and humid climates. The project demonstrated a strong combination of engineering rigor, user-centric design, and sustainability thinking. The team undertook a structured process involving ideation, user research, prototype development, and real-world testing, leading to a solution that is both technically sound and socially relevant.

Ryan contributed significantly to the technical development and testing of the prototype, which integrated thermoelectric cooling with desiccant-based dehumidification. The system was designed to be compact, energy-efficient, and easily installable on existing ceiling fans, making it particularly suitable for low- and middle-income households.

The team's work was validated through controlled testing in real residential settings, where the prototype demonstrated substantial energy savings compared to conventional air-conditioning systems, along with improved thermal comfort and reduced humidity levels. Notably, the solution showed the potential to reduce energy consumption by a significant margin while maintaining comfortable indoor conditions, highlighting both its technical robustness and scalability.

Beyond technical execution, Ryan also demonstrated the ability to work effectively in a multidisciplinary team, contributing to aspects such as design development, problem-solving, and iterative refinement based on user feedback. His work reflects a strong

understanding of the intersection between engineering, sustainability, and real-world applicability.

Overall, Ryan is a motivated and capable individual with a solid foundation in engineering design and a clear commitment to developing practical, climate-responsive solutions. I am confident that he will be a valuable addition to any academic or professional environment he joins.

Sincerely



Prasad Vaidya
Director, Solar Decathlon India